

PLCA vs CSMA/CD on one 10BASE-T1S segment, repeated

Generated by `make_access_report.py` from `access_20261006_114158.json`.

Bench. ESP-A: ESP32-S3 + LAN8651 MAC-PHY, PLCA coordinator (id 0 of 2), SPI 20.00 MHz. Converter: LAN8670 + LAN9355, dial id 1. ESP-B: ESP32-S3 + W5500 on the converter's 100BASE-TX port, in the PC's place. 1000 B UDP. The same five phases as the 2026-10-01 PC comparison; every number below is per round, mean $\pm$ 95 % CI (Student t) over rounds.

CSMA/CD rounds pending. The converter's PLCA is a DIP switch. With it on and only the node switched to `csma`, the converter does not fall back to CSMA/CD: it stops transmitting (every probe and every flood datagram from ESP-B lost, `access_20261006_113637`). The CSMA/CD rounds need DIP 1 off by hand; this report fills its second column when they exist.

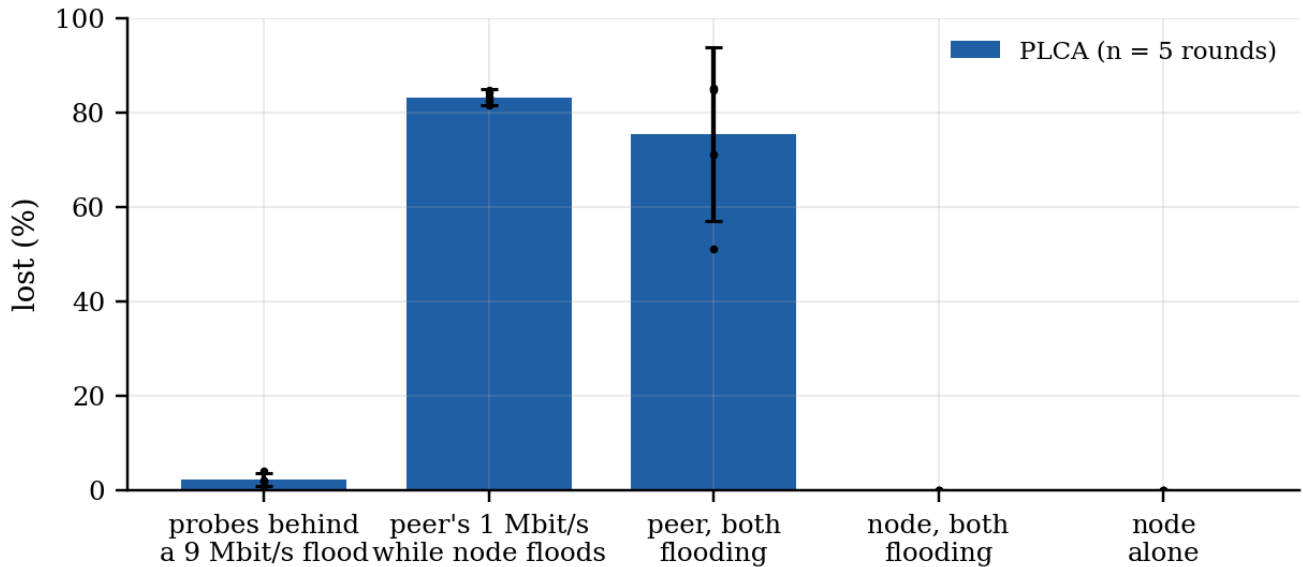


Fig. A1 — Loss in each phase, per method; bars are means, whiskers 95 % CI, dots single rounds.

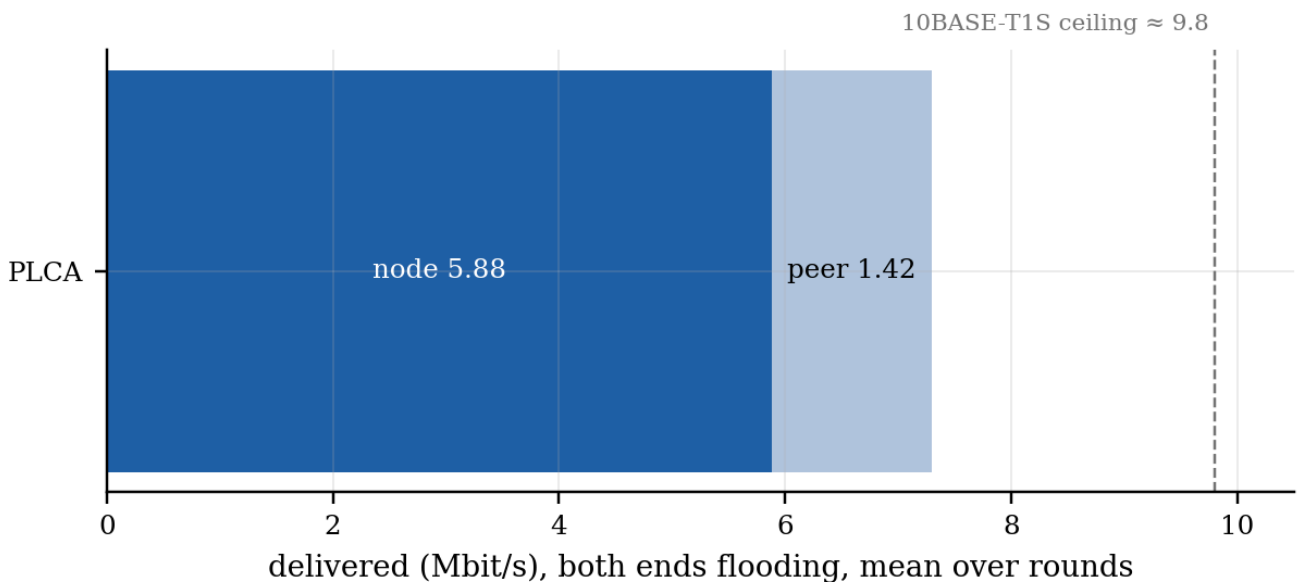


Fig. A2 — How the bus is shared when both ends flood: the node's delivered rate (solid) and the peer's (light).

phase	metric	PLCA (n = 5)
idle	64 B round trip, median (from ESP-B)	2.59 ± 0.01 ms (2.58–2.60)
idle	probes lost	0.0 ± 0.0 % (0.0–0.0)
peer floods 9 Mbit/s	node's probes lost	2.2 ± 1.4 % (1.0–4.0)
peer floods 9 Mbit/s	node's probe round trip, median	3.04 ± 0.19 ms (2.86–3.23)
peer floods 9 Mbit/s	node's probe round trip, p99	1730.18 ± 1041.21 ms (821.54–2639.82)
peer floods 9 Mbit/s	flood delivered to the node	8.52 ± 0.02 Mbit/s (8.51–8.54)
peer floods 9 Mbit/s	flood lost	5.3 ± 0.2 % (5.1–5.5)
node floods, peer 1 Mbit/s	node -> peer delivered	7.05 ± 0.01 Mbit/s (7.04–7.06)
node floods, peer 1 Mbit/s	node -> peer lost	0.00 ± 0.00 % (0.00–0.00)
node floods, peer 1 Mbit/s	peer -> node delivered (of 1.0)	0.17 ± 0.02 Mbit/s (0.15–0.18)
node floods, peer 1 Mbit/s	peer -> node lost	83.2 ± 1.7 % (81.4–84.6)
both flood	node -> peer delivered	5.88 ± 1.10 Mbit/s (4.49–6.48)
both flood	node -> peer lost	0.00 ± 0.00 % (0.00–0.00)
both flood	node's longest gap	527 ± 977 ms (7–1782)
both flood	peer -> node delivered	1.42 ± 1.30 Mbit/s (0.73–3.13)
both flood	peer -> node lost	75.4 ± 18.4 % (51.0–85.1)
node alone	node -> peer delivered	7.22 ± 0.01 Mbit/s (7.22–7.23)
node alone	node -> peer lost	0.00 ± 0.00 % (0.00–0.00)
node alone	arrival gap p99	2.50 ± 0.00 ms (2.50–2.50)

Reading it (PLCA). The node never lost a datagram it sent: 0.00 % while it flooded alone, 0.00 % with the peer sending, 0.00 % with both flooding. Its small probes got onto a bus the peer was flooding at 9 Mbit/s with 2.2 % loss (median 3.04 ms), but their tail is in seconds (p99 1.73 s): the peer offers 9 Mbit/s and the converter carries 8.52 onto T1S, so a queue builds on that path and the replies wait behind it — the median hides it. What PLCA does not protect is the converter's own stream: while the node floods, 83 % of the peer's 1 Mbit/s is lost, and with both flooding 75 % — the converter's two-way limitation the vendor lists (issue 4.3), seen in every round. When both flood, the node keeps 5.88 Mbit/s but stalls for up to 527 ms (mean of per-round maxima): it waits for transmit opportunities rather than losing frames.

Limits. Two nodes on the bus, one of them a converter whose transmit path is the known weak point; rounds of one method run back to back (the DIP is set by hand), so slow drift between the two blocks is not randomised out; one bus length (~1 m).